

[Aim: 100/100 in Maths]

आरंभ

POLYNOMIALS

Lecture #3

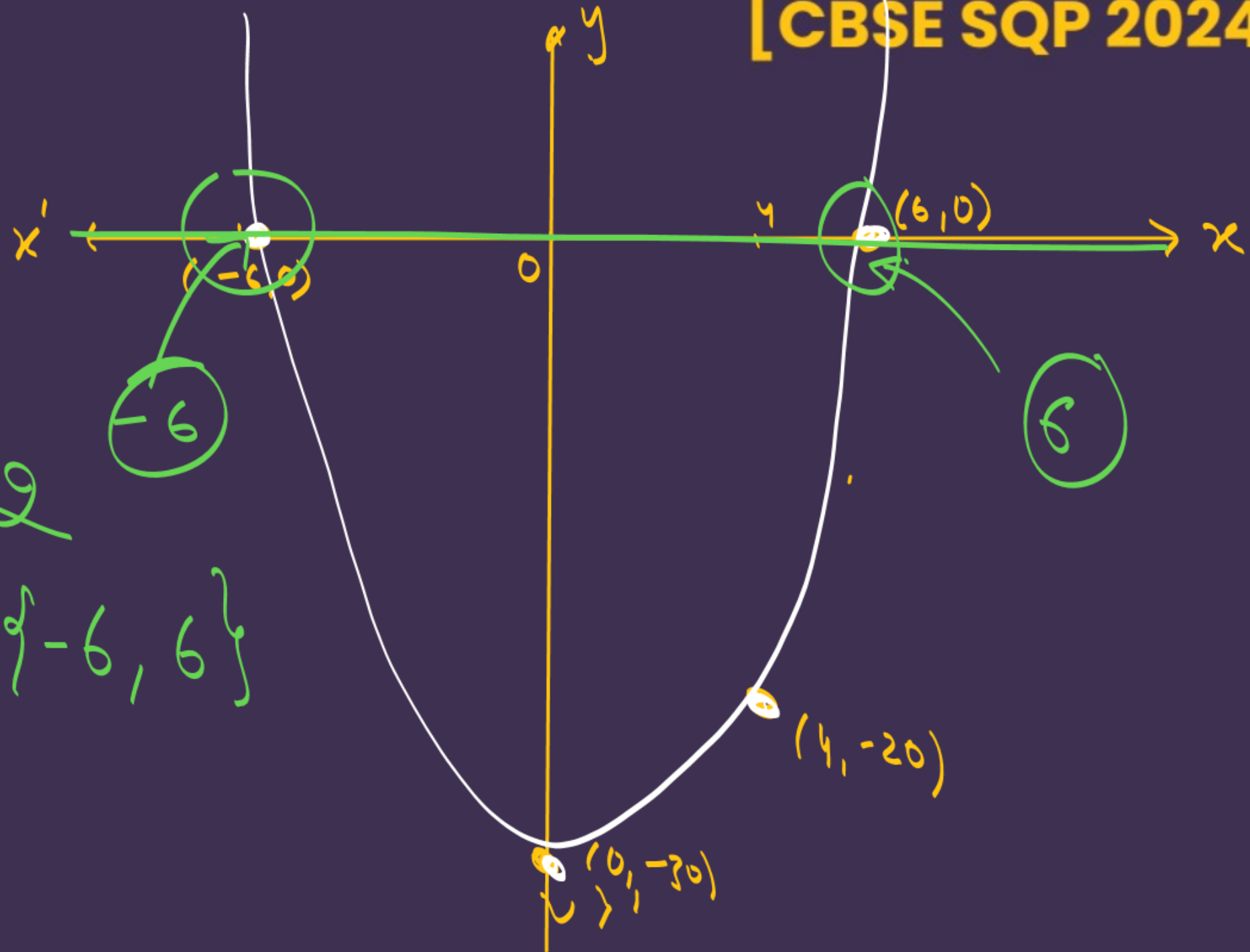
#LP: The graph of quadratic polynomial $p(x)$ passes through the points $(-6, 0)$, $(0, -30)$, $(4, -20)$ and $(6, 0)$. The zeroes of the polynomial are: [CBSE SQP 2024-25]

(A) $-6, 0$

(B) $4, 6$

(C) $-30, -20$

(D) $-6, 6$



no. of zeroes = 2
zeroes value $\Rightarrow \{-6, 6\}$

#LP: Assertion (A): If the graph of a polynomial intersects the x-axis at exactly two points, then the number of zeroes of that polynomial is 2. (T)

Reason (R): The number of zeroes of a polynomial is equal to the number of points where the graph of the polynomial intersects x-axis. (T)

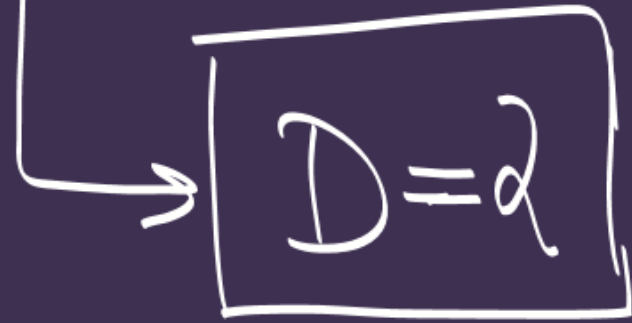
[CBSE Comptt., 2024]

A ✓
R ✓

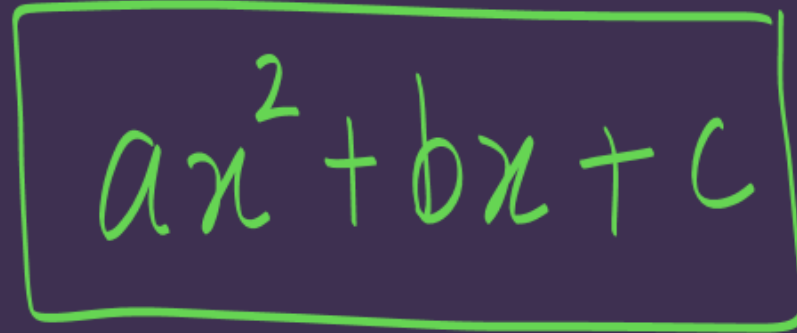
R is correct exp. of A ✓

(A) ✓

Quadratic Polynomial:


$$D=2$$

$\Rightarrow$


$$ax^2 + bx + c$$

Middle Term Splitting Method to find Zero of a Quadratic Polynomial:

$$x^2 + 10x + 24 = 0$$

$$x^2 + 4x + 6x + 24 = 0$$

$$x(x+4) + 6(x+4) = 0$$

$$(x+4)(x+6) = 0$$

$$x+4=0$$

$$x = -4$$

$$x+6=0$$

$$x = -6$$

$$p+q=10 \checkmark$$

$$p \times q = 24$$

$$\begin{array}{r|l} 2 & 24 \\ \hline 4 & 12 \end{array}$$

$$\begin{array}{r|l} 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$$3x^2 - 11x - 4 = 0$$

$$3x^2 - 12x + 1x - 4 = 0$$

$$3x(x-4) + 1(x-4) = 0$$

$$(x-4)(3x+1) = 0$$

$$x-4=0$$

$$x = 4$$

$$3x+1=0$$

$$x = -\frac{1}{3}$$

$$p+q=-11$$

$$p \times q = -12$$

#LP: find the zeroes $\rightarrow x^2 + x - 6 = 0$

$$\underbrace{x^2 - 2x} + \underbrace{3x - 6} = 0$$

$$\underline{x}(x-2) + \underline{3}(x-2) = 0$$

$$(x-2)[x+3] = 0$$

$$x-2=0$$

$$x=2$$

$$x+3=0$$

$$x=-3$$

$$p+q=1$$

$$p \times q = -6$$

2	6
3	3
	1

#LP: Find the roots of $3x^2 - 27$.

$$3x^2 - 27 = 0$$

$$3(x^2 - 9) = 0$$

$$x^2 - 9 = 0$$

$$x^2 = 9$$

$$x = \sqrt{9}$$

$$x = \pm 3$$

$$(x)^2 - (3)^2 = 0$$

$$(x+3)(x-3) = 0$$

$$x+3=0$$

$$x = -3$$

$$x-3=0$$

$$x = +3$$

$$x^2 = 4$$

$$x = \sqrt{4}$$

$$x = \pm 2$$

$$x^2 - 4 = 0$$

$$x^2 - (2)^2 = 0$$

$$(x+2)(x-2) = 0$$

$$x+2=0 \quad | \quad x-2=0$$

$$x = -2 \quad | \quad x = +2$$

#LP: Find the roots of $x^2 - 11/6 x - 5/3$

$$\Rightarrow x^2 - \frac{11}{6}x - \frac{5}{3} = 0$$

$$L(M) \Rightarrow \frac{6x^2 - 11x - 10}{6} = 0$$

$$\Rightarrow \boxed{6x^2 - 11x - 10 = 0}$$

$$\boxed{6x^2 + 4x - 15x - 10 = 0}$$

$$\underline{2x} (3x + 2) - \underline{5} (3x + 2) = 0$$

$$(3x + 2) [2x - 5] = 0$$

$$p + q = -11$$
$$p \times q = -60$$

$$3x + 2 = 0 \quad 2x - 5 = 0$$
$$x = -2/3 \quad x = 5/2$$

4	2	60
	2	30
15	3	15
	5	5
		1

~~#K3B~~

$D=1 \longrightarrow$ max one zero

$D=\underline{2} \longrightarrow$ max two zeroes

$D=3 \longrightarrow$ max three zeroes

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$D=2$, no of zero

no zero ✓

one zero ✓

two zeroes ✓

three zeroes ✓

#LP: Assertion (A): The polynomial $p(x) = x^2 + 3x + 3$ has two real zeroes.

Reason (R): A quadratic polynomial can have at most two real zeroes. **[CBSE 2023]**

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**
- (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).**
- (C) Assertion (A) is true, but Reason (R) is false.**
- (D) Assertion (A) is false, but Reason (R) is true.**

HW

Module $\rightarrow$ $1P(I) \rightarrow 1, 2, 3, 4$

$1P(II) \rightarrow 3$

$2P(I) \rightarrow 1, 5, 1^2, 1^3$

THANK YOU COODIESS !!

